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How to Train Your AI Dragon: Email #2 - Literature Synthesis

Hello again! Welcome to email #2 of this short series for social scientists. (Was this forwarded? Sign up here).

Today, we're talking about how to use AI for literature synthesis. But first, I wanted to share that I finally watched How to Train Your Dragon. Interestingly, the different versions of the movie work as another AI analogy.

The original (2010) and the sequels (2014, 2019) were all animated, while the latest version (2025) is live action.

The live action version is much closer to real life, but you still know it's a movie. Just like how LLMs are much closer to sounding human than ever before, but they are still just computer programs.

Now back to training your AI dragon on searching the literature. Imagine you've just been asked to brief a team on ethnic variation in Parkinson's disease (PD) prevalence across Australasia. It's not your primary field. You're short on time, the topic is complex, and you don't want to miss the key debates and data gaps.This is exactly where AI can shine.

The Hiccup Model: Observe Before You Act

In *How to Train Your Dragon*, our hero Hiccup survives because he observes before he acts. While others are busy charging in, he watches the dragon, notes its reactions, and only then starts experimenting.

Working with AI to search the literature is the same. If you rush in and treat outputs as instant facts, you're essentially charging a dragon head-on. The tool will give you fluent, plausible answers even when the evidence is shallow, outdated, or nonexistent. That's how you get confidently wrong statements about niche topics.

Instead, use AI as an observation tool:

- Observe the landscape: What are the main hypotheses about ethnic variation in PD in this region?
- Observe the fault lines: Where do studies disagree, and why?
- Observe the blind spots: Which groups, like Māori or Pasifika, are barely represented in the data?

You're asking, "What are people talking about, and what might I be missing?"

Building Your "Dragon's Hoard" Reference Database

In my own work, I've found that training your AI dragon is much easier if you give it a structured environment. Instead of a static list in EndNote, I use an AI-assisted workspace to organize references by year, population, and topic in a table I can query. I can also add to it automatically, with an autonomous search agent that pulls relevant papers as they're published and adds them to the database with a link.

A stack of papers that I used to keep in a reference manager is now a dynamic body of literature I can sort, filter, and search in seconds. If I need references exported in any number of journal styles, I just tell the tool what I want – no more hunting around looking for styles to manually import!

The Verification Imperative

Unlike a human expert, an LLM has no internal alarm bell that rings when it wanders into thin evidence. It can't tell you what it doesn't know.

This is why I suggest the following guardrails:

- **Verify every number.** Never trust a quoted prevalence rate or p-value without checking the primary source.
- **Audit citations.** AI can hallucinate references. If you haven't opened the PDF yourself, don't cite it.
- **Be transparent.** Disclose your use of AI tools to co-authors and editors.

A Practical, Step-by-Step Workflow

1. **Frame the Question Precisely:** Don't ask, "Tell me about PD." Be specific: *"Summarize major findings on ethnic variation in PD in Australasia since 2000. Emphasize Māori and Pasifika populations. What methodological issues complicate the data? Cite your sources."*
2. **Get the Map, Not the Verdict:** Use AI to generate a whiteboard-style brainstorm of key themes, cohorts, and recurrent caveats, like diagnostic bias.
3. **Cross-Check with Real Databases:** Take your refined queries to PubMed or Google Scholar. The AI can help you construct Boolean searches. Use the concepts you surfaced to build better, sharper search terms. This is where you build your real understanding.
4. **Synthesize with Human Judgment:** You remain the final arbiter. You decide which studies are weak and where the real uncertainty lies. AI can help you write faster, but it can't substitute for your professional judgment.

The Bottom Line

Used well, AI turns "I have no idea where to start" into "I have a plan" in a fraction of the time. But speed is a trap if you skip the verification. Be like Hiccup: lead with curiosity, but keep your eyes open.

To push your own practice: Where in your current workflow are you most tempted to skip verification, and what would make it easier to stay disciplined?

Next week, we'll dive into Grant Writing. See you then!

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